

Basic Modular Tensor Category

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April 25, 2026

Abstract

This is a note on the topic of Modular Tensor Categories. It is mainly based on the discussion in [1]

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1 Preliminaries

This section we introduce some concepts in Algebra and Representation that may help us understand category as a more general mathematical structure.

1.1 Intertwiner and Representation

Let G be a group and let (V, ρ) , (W, σ) be two representations of G . A **intertwiner** (or **morphism of representations**) is a linear map

$$\varphi : V \rightarrow W \tag{1}$$

such that for all $g \in G$,

$$\varphi \circ \rho(g) = \sigma(g) \circ \varphi, \tag{2}$$

i.e., the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\rho(g)} & V \\ \varphi \downarrow & & \downarrow \varphi \\ W & \xrightarrow{\sigma(g)} & W \end{array} \tag{3}$$

The set of all intertwiners from (V, ρ) to (W, σ) is denoted $\text{Hom}_G(V, W)$. If φ is an isomorphism, the two representations are said to be **equivalent** (or **isomorphic**), written $(V, \rho) \simeq (W, \sigma)$.

Theorem 1.1**Schur's Lemma**

Let (V, ρ) and (W, σ) be irreducible representations of G , and let $\varphi \in \text{Hom}_G(V, W)$.

1. If $(V, \rho) \not\cong (W, \sigma)$, then $\varphi = 0$.
2. If $V = W$ and G acts over $\mathbb{C}$, then $\varphi = \lambda \text{id}_V$ for some $\lambda \in \mathbb{C}$.

This is a version of Shur's Lemma using the language of intertwiners.

1.1.1 Intertwiners of Lie Algebras

Now let $\mathfrak{g}$ be a Lie algebra and let $(V, \rho), (W, \sigma)$ be representations of $\mathfrak{g}$, i.e., Lie algebra homomorphisms

$$\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V), \quad \sigma : \mathfrak{g} \rightarrow \mathfrak{gl}(W). \quad (4)$$

A **intertwiner of Lie algebra representations** is a linear map $\varphi : V \rightarrow W$ satisfying

$$\varphi \circ \rho(X) = \sigma(X) \circ \varphi, \quad \forall X \in \mathfrak{g}. \quad (5)$$

Equivalently, φ intertwines the $\mathfrak{g}$ -actions:

$$\varphi(X \cdot v) = X \cdot \varphi(v), \quad \forall X \in \mathfrak{g}, v \in V, \quad (6)$$

where $X \cdot v := \rho(X)(v)$.

The category of representations of $\mathfrak{g}$ with intertwiners as morphisms is denoted $\text{Rep}(\mathfrak{g})$.

Example

Let $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ with standard generators e, f, h satisfying $[h, e] = 2e, [h, f] = -2f, [e, f] = h$. The finite-dimensional irreducible representations V_n ($n \in \mathbb{N}$) have basis $\{v_0, v_1, \dots, v_n\}$ with

$$h \cdot v_k = (n - 2k)v_k, \quad e \cdot v_k = (n - k + 1)v_{k-1}, \quad f \cdot v_k = (k + 1)v_{k+1}. \quad (7)$$

By Schur's Lemma, $\text{Hom}_{\mathfrak{sl}_2}(V_m, V_n) = 0$ for $m \neq n$, and $\text{End}_{\mathfrak{sl}_2}(V_n) \simeq \mathbb{C}$.

1.1.2 CVO as Intertwiners

The Chiral Vertex Operators constructed in Moore and Seiberg's work can be understood as king of a generalization of the concept of intertwiners of representation of Virasoro Algebra (in fact any Chiral Algebra(eg. Kac Moody Algebra) can be used here).

See [2] for a constructive definition of CVOs. In the constructive definition, we are made believe that the CVOs are defined to compute the BPZ conformal blocks of RCFT (at least for ones on a plane and on a torus).

There is also an axiomatic definition of CVOs as a intertwining object between:

$$\begin{pmatrix} i \\ jk \end{pmatrix}(z) : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A} \quad (8)$$

where $\mathcal{A}$ is a representation of the chiral algebra of CFT, eg. Virasoro Algebra.

However, instead of following Equation 6 as a normal lie algebra, it satisfy a more complicated intertwining relation:

- We first define a “co-product” of the chiral algebra representation:

$$\Delta_z : \mathcal{A} \rightarrow \mathcal{A} \otimes \mathcal{A}, \quad \Delta \quad (9)$$

2 Modular Tensor Categories

2.1 Category Basics

A Category $\mathcal{C}$ is a mathematical structure with 2 type of data:

- Objects: $\text{Obj}(\mathcal{C})$, for example, $U, V \in \text{Obj}(\mathcal{C})$.
- Morphisms: for any two objects $U, V \in \text{Obj}(\mathcal{C})$, there is a set of morphisms $\text{Hom}(U, V)$, for example, $f \in \text{Hom}(U, V)$.

In the CFT realization of a modular tensor category, one understands the object as

Bibliography

- [1] J. Fuchs, I. Runkel, and C. Schweigert, “TFT Construction of RCFT Correlators I: Partition Functions,” *Nuclear Physics B*, vol. 646, no. 3, pp. 353–497, Dec. 2002, doi: [10.1016/S0550-3213\(02\)00744-7](https://doi.org/10.1016/S0550-3213(02)00744-7).
- [2] G. W. Moore and N. Seiberg, “Lectures on RCFT,” in *1989 Banff NATO ASI: Physics, Geometry and Topology*, 1989, pp. 1–129.